



File Handling

Practical Skills Module 4: Topic 5

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Topic Code	PM-04-PS04
Topic	Handle files

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0401 Open a text file
- PA0402 Create a text file
- PA0403 Append data to an existing file
- PA0404 Read the ENTIRE contents of a file
- PA0405 Read the content of the file line by line
- PA0406 Write a text file
- PA0407 Check if a file or directory exist
- PA0408 Copy files
- PA0409 Rename a file and directory
- PA0410 Create a Zip file
- PA0411 Write and read a binary file

Applied Knowledge

- AK0401 Concept, definition and functions of intermediate Python programming principles

Internal Assessment Criteria

- IAC0401 Expected results with file handling in Python are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-04



Practical task

Follow the facilitator's instructions to complete the following activities:

Step 1: Create and Open a Text File

Scenario: The business needs to prepare customer records for review.

- Create a file named `customers.txt` using **write ('w') mode**.

Add the following example customer data to the file:

```
1001, Alice Smith, alice@example.com
1002, Bob Jones, bob@example.com
1003, Clara Patel, clara@example.com
```

- Close the file.
- Then, reopen the file in **read ('r') mode** to verify its content by printing it.

Step 2: Create a Text File for Inventory

Scenario: Generate a new file to store the day's inventory items.

Instructions (in Python):

- Create a new file called `inventory.txt`.

Add the following inventory items:

```
Item001, Laptop, 5  
Item002, Mouse, 25  
Item003, Keyboard, 10
```

Expected Result:

A new file is created and filled with the given data.

Step 3: Append Data to the Customer File

Scenario: A new customer has registered and needs to be added to the record.

- Reopen `customers.txt` in **append ('a') mode**.

Add the following line:

```
1004, Daniel Lee, daniel@example.com
```

- Read and print all lines again to confirm the data was added.

Step 4: Read and Display Full File Content

Scenario: Display the full inventory for an end-of-day report.

- Open `inventory.txt` in **read ('r') mode**.
- Use `read()` to display the entire content in one go.

Step 5: Read the File Line-by-Line

Scenario: Process customer records one at a time.

- Use a loop to read each line of `customers.txt` and print the name from each record.

Hint: You can split each line using `.split(",")` to extract the name.

Step 6: Create a Monthly Sales Report File

Scenario: Write a new text file to store sales totals for each week of the month.

Create `monthly_sales.txt` and write the following lines:

```
Week 1: R12000
Week 2: R15000
Week 3: R14000
Week 4: R16000
```

Step 7: Check if a Backup File Exists

Scenario: Check for a file named `customers_backup.txt`.

- Use `os.path.exists()` or `pathlib.Path.exists()` to check.
- Print a message stating whether it exists.

Step 8: Copy the Customer File

Scenario: Create a backup of the customer records.

- Use the `shutil` module to copy `customers.txt` to `customers_backup.txt`.

Step 9: Rename the Inventory File

Scenario: Update the inventory file's name to include today's date.

- Rename `inventory.txt` to `inventory_2025_05_23.txt` using `os.rename()`.

Step 10: Create a Zip File

Scenario: Compress both customer and inventory files into a zip archive.

- Use the `zipfile` module to zip:
 - `customers.txt`
 - `inventory_2025_05_23.txt`

- Name the archive `daily_backup.zip`.

Step 11: Write and Read a Binary File

Scenario: Save and retrieve a profile picture file (simulated binary file).

Create a small binary file by writing some bytes:

```
with open('profile_image.bin', 'wb') as bin_file:  
    bin_file.write(b'\x89PNG\r\n\x1a\n')
```

- Then read and print the binary data using `'rb'`.



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
